Infection Segmentation Using U-Net

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I. Introduction

In this project, we use the COVID-QU-Ex Dataset from Kaggle platform, collected by researchers of Qatar University. The dataset consists of 33,920 chest X-ray images:

- COVID-19 cases: 11,956
- Non-COVID infections: 11,263
- Normal cases: 10,701 images

This is the largest ever created lung mask dataset and also the first study that utilizes both lung and infection segmentation to localize and quantify COVID-19 infection from X-ray images. Therefore, it can assist medical doctors in better diagnosing the severity of COVID-19 and easily following up the progression of the disease.

The experiments were conducted on two CXR sets, where each set is divided into training, validation, and test sets:

- Lung Segmentation Data
- COVID-19 Infection Segmentation

In this work, we only focus on infection segmentation and ignore lung mask segmentation.

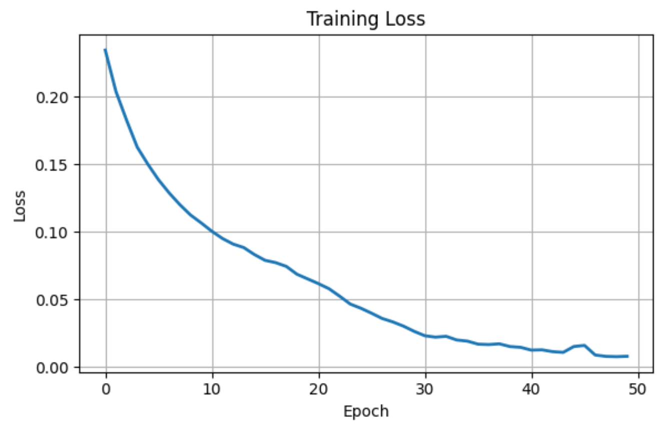
II. U-Net Model

We implement a custom `load_dataset` function to load chest X-ray images and their corresponding infection segmentation masks for three categories: COVID-19, Non-COVID infections, and Normal cases. All images are already in grayscale format and are normalized to the range [0,1].

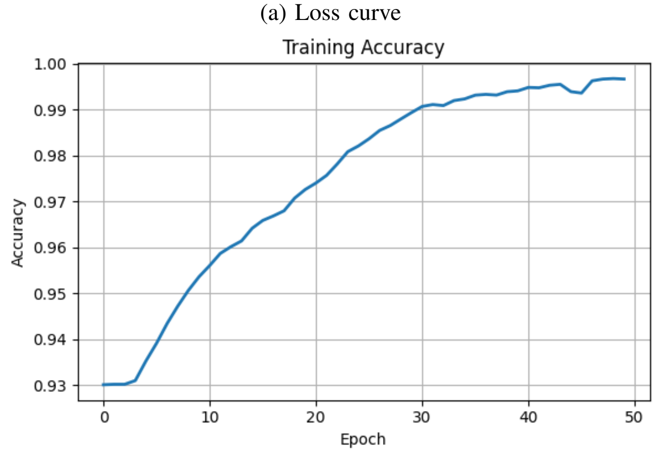
The ground-truth masks are processed into binary segmentation masks, where infected regions are labeled as 1 and background regions as 0. A channel dimension is added to match the input format required by convolutional neural networks. The processed image–mask pairs are then used for training and evaluating the segmentation model.

We build a custom U-Net model consisting of three main components: encoder, bottleneck, and decoder. The encoder contains three convolutional blocks, each followed by a max-pooling layer to reduce spatial resolution while extracting high-level features. A bottleneck block is used to capture the most abstract representations of the input images. The decoder upsamples the feature maps and concatenates them with the corresponding encoder features. Finally, a sigmoid activation function is applied to produce a binary segmentation mask.

III. Result



(a) Loss curve



(b) Accuracy curve

Fig. 1: Training accuracy and Loss curve

From the graph, we clearly see that the model achieved 99% accuracy and only 1% loss. However, this does not illustrate that the model performs well as infection takes a small amount of pixels. For example, 256x256 image contains 65,000 pixels with only about 2000 infections pixels and remaining 63000 are background, if the model predict all of pixels are background, we will get 63000/65000 = 97% but missing all the infection (which is the most information that we have to predict).

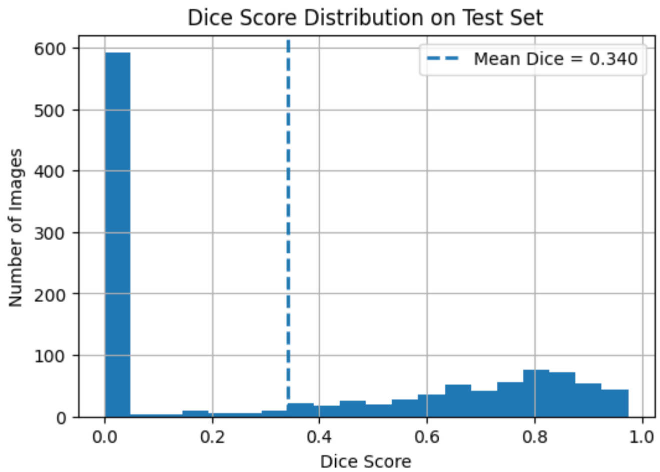


Fig. 2: Dice Score Distribution

For segmentation tasks, we use Dice Score, which measured by calculating the overlap between predicted and the ground-truth pixels. The figure shows that the median Dice score is quite low (0,34), this is caused by many of input images do not contain infection, making the model predict 0, leading to low median dice score.

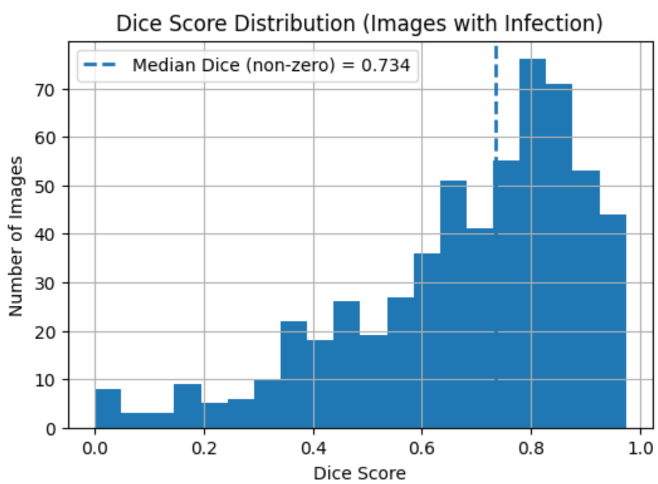


Fig. 3: Dice Score Distribution (Images with Infections)

The figure illustrates that for infections images, the model perform well with median Dice score is 0.73, making it reliable for infection segmentation tasks.

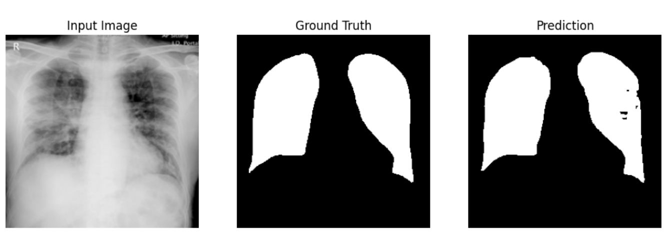


Fig. 4: Example of model result

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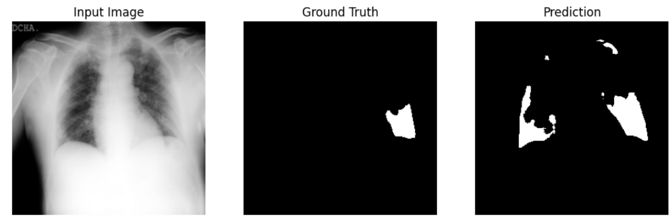

Fig. 5: Example of model result

Two figures present qualitative segmentation results produced by the model.